

File Information Product : T FLONOXIN 400MG TAB SAP No. : Version : RASUB/V1 Date : 29 May 2024		Artwork Type [PLEASE TICK ✓] ☐ Unit Box ☒ Leaflet ☐ Aluminium Foil ☐ Outer Box ☐ Fix-A-Form ☐ Sachet ☐ Label ☐ Others : _____		Sign / Stamp	
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Flonoxin 400mg tablet

DESCRIPTION:
Flonoxin 400mg tablet: Dull red, 17 mm oblong shaped, plain on both sides, film-coated tablet.

COMPOSITION:
Each film coated tablet contains 400 mg Moxifloxacin.

PHARMACODYNAMICS:
Moxifloxacin is a DNA gyrase inhibitor and also inhibits topoisomerase IV. DNA gyrase (topoisomerase II) is an essential bacterial enzyme that maintains the superhelical structure of DNA. DNA gyrase is required for DNA replication and transcription, DNA repair, recombination and transposition; inhibition is bactericidal.

PHARMACOKINETICS:
Moxifloxacin is well-absorbed with a bioavailability of 90%. Its absorption is not affected by high-fat meal or yogurt. The volume of distribution is approximately 1.7-2.7 L/kg with tissue concentration often exceeding plasma concentrations in respiratory tissues, alveolar macrophages, abdominal tissues/ fluids, uterine tissue (endometrium, myometrium), and sinus tissues. Moxifloxacin is approximately 30-50% bound to proteins. 52% of the dose is hepatically metabolised via glucuronide and sulfate conjugation. The half-life elimination of a single oral dose is 12-16 hours. Moxifloxacin is excreted via urine (as unchanged druged and glucuronide conjugates) and feces (as unchanged drug and sulfate conjugates).

INDICATIONS:
Moxifloxacin tablets are indicated for treatment of adults (≥18 years of age) with the following bacterial infections caused by susceptible strains:
• Acute bacterial rhinosinusitis*
• Acute exacerbations of chronic obstructive pulmonary disease including chronic bronchitis*
• Nosocomial pneumonia/Hospital-acquired pneumonia*
• Acute otitis media*
• External otitis*
• Endocarditis*
• Infection of cerebrospinal fluid*
• Meningitis*
• Septicaemia*
• Uncomplicated acute cystitis / uncomplicated cystitis*
• Prevention of exacerbations in women with recurring urinary tract infections*
• Prevention of infection in surgical procedures in the urogenital system*,#
• Pre-operative preparations for chronic cholesteatomatous otitis and chronic otitis spreading to bone*
• Community acquired pneumonia
• Mild to moderately severe inflammatory pelvic diseases (i.e. infections of the upper female genital tract, including salpingitis and endometritis) without an associated tubo-ovarian or pelvic abscess. Moxifloxacin is not recommended for monotherapy of mild to moderately severe inflammatory pelvic diseases. Preferably, they should be administered in combination with another suited antibiotic (eg: cephalosporin), due to the increasing resistance of Neisseria gonorrhoeae to moxifloxacin; that is, unless moxifloxacin-resistant Neisseria gonorrhoeae can be ruled out.
• Complicated skin and skin structure infections
• Complicated intraabdominal infections including polymicrobial such as abscesses
Consideration should be given to official guidance on the appropriate use of antibacterial agents.

*Flonoxin should be only used :
• When Pseudomonas is considered AND the patient is allergic to antipseudomonal penicillins/ cephalosporins;
• For resistant organisms with no other alternative antibiotics available.
#Flonoxin should not be used >24 hours post operation.

RECOMMENDED DOSAGE:
Dose
Adult: One 400 mg film-coated tablet once daily
Pediatrics: The efficacy of Moxifloxacin in children and adolescents has not been established. No recommendation on dosology can be made. The safety of Moxifloxacin in children below the age of 6 years has not been established.
Renal impairment: No adjustment of dosage is required in patients with mild to severely impaired renal function or in patients on chronic dialysis i.e. haemodialysis and continuous ambulatory peritoneal dialysis.
Hepatic impairment: No dosage adjustment is required in patients with impaired liver function.
Other special populations: No adjustment of dosage is required in the elderly and in patients with low bodyweight

Duration
The duration of treatment should be determined by the severity of the indication or clinical response. Moxifloxacin tablets should be used for the following treatment durations:
• Acute exacerbation of chronic bronchitis: 5 days
• Community acquired pneumonia: 10 days
• Acute bacterial sinusitis: 7 days
• Mild to moderate pelvic inflammatory disease: 14 days
• Complicated skin and skin structure infections total treatment duration for sequential therapy (intravenous followed by oral therapy): 7-21 days
• Complicated intraabdominal infections total treatment duration for sequential therapy (intravenous followed by oral therapy): 5-14 days

The recommended dose and duration of therapy for the indication being treated should not be exceeded.

ROUTE OF ADMINISTRATION:
For oral administration only. The film-coated tablet should be swallowed whole with sufficient liquid and may be taken independent of meals.

CONTRAINDICATIONS:
• Hypersensitivity to moxifloxacin, other quinolones or to any of the excipients
• Pregnancy and lactation
• Patients below 18 years of age
• Patients with a history of tendon disease/ disorder related to quinolone treatment
Changes in cardiac electrophysiology have been observed following exposure to moxifloxacin, in the form of QT prolongation. For reasons of drug safety, moxifloxacin is therefore contraindicated in patients with:
• Congenital or documented acquired QT prolongation
• Electrolyte disturbances, particularly in uncorrected hypokalaemia
• Clinically relevant bradycardia
• Clinically relevant heart failure with reduced left-ventricular ejection fraction
• Previous history of symptomatic arrhythmias
Moxifloxacin should not be used concurrently with other drugs that prolong the QT interval.
Due to limited data, moxifloxacin is also contraindicated in patients with impaired liver function (Child Pugh C) and in patients with transaminase increase > 5fold ULN.

WARNINGS AND PRECAUTIONS:
The use of moxifloxacin should be avoided in patients who have experienced serious adverse reactions in the past when using fluoroquinolones containing products (see section Adverse Effects/Undesirable Effects). Treatment of these patients with moxifloxacin should only be initiated in the absence of alternative treatment options and after careful benefit/risk assessment.

Prolonged, disabling and potentially irreversible serious adverse drug reactions
Very rare cases of prolonged (continuing months or years), disabling and potentially irreversible serious adverse drug reactions affecting different, sometimes multiple body systems (musculoskeletal, nervous, psychiatric and senses) have been reported in patients receiving fluoroquinolones irrespective of their age and pre-existing risk factors. Moxifloxacin should be discontinued immediately at the first signs or symptoms of any serious adverse reaction and patients should be advised to contact their prescriber for advice.

Prolongation of QTc interval and potentially QTc-prolongation-related clinical conditions
Moxifloxacin has been shown to prolong the QTc interval on the electrocardiogram in some patients. As women tend to have a longer baseline QTc interval compared with men, they may be more sensitive to QTc-prolonging medications. Elderly patients may also be more susceptible to drug-associated effects on the QT interval. Moxifloxacin should be used with caution in patients with ongoing proarrhythmic conditions (especially women and elderly patients), such as acute myocardial ischaemia or QT prolongation as this may lead to an increased risk for ventricular arrhythmias (including torsade de pointes) and cardiac arrest.

The magnitude of QT prolongation may increase with increasing concentrations of the drug. Therefore, the recommended dose should not be exceeded. If signs of cardiac arrhythmia occur during treatment with moxifloxacin, treatment should be stopped and an ECG should be performed.

Hypersensitivity/ allergic reactions
Hypersensitivity and allergic reactions have been reported for fluoroquinolones including moxifloxacin after first administration. Anaphylactic reactions can progress to a life-threatening shock, even after the first administration. In cases of clinical manifestations of severe hypersensitivity reactions, moxifloxacin should be discontinued and suitable treatment (eg: treatment for shock) initiated.

Severe liver disorders
Cases of fulminant hepatitis potentially leading to liver failure (including fatal cases) have been reported with moxifloxacin. Patients should be advised to contact their doctor prior to continuing treatment if signs and symptoms of fulminant hepatic disease develop such as rapidly developing asthenia associated with jaundice, dark urine, bleeding tendency or hepatic encephalopathy. Liver function tests/ investigations should be performed in cases where indications of liver dysfunction occur.

Serious bullous skin reactions
Cases of bullous skin reactions like Stevens-Johnson syndrome or toxic epidermal necrolysis have been reported with moxifloxacin. Patients should be advised to contact their doctor immediately prior to continuing treatment if skin and/or mucosal reactions occur.

Patients predisposed to seizures
Quinolones are known to trigger seizures. Use should be with caution in patients with CNS disorders or in the presence of other risk factors which may predispose to seizures or lower the seizure threshold. In case of seizures, treatment with moxifloxacin should be discontinued and appropriate measures instituted.

Peripheral neuropathy
Cases of sensory or sensorimotor polyneuropathy resulting in paraesthesias, hypoaesthesias, dysaesthesias, or weakness have been reported in patients receiving quinolones including moxifloxacin. Patients under treatment with moxifloxacin should be advised to inform their doctor prior to continuing treatment if symptoms of neuropathy such as pain, burning, tingling, numbness or weakness develop in order to prevent the development of an irreversible condition. See section Adverse Effects/Undesirable Effects.

Psychiatric reactions
Psychiatric reactions may occur even after the first administration of fluoroquinolones, including Flonoxin 400mg Tablet. In rare cases, depression or psychotic reactions can progress to suicidal ideations/thoughts and self-injurious behaviour, such as attempted or completed suicide (see section 'Undesirable effects'). In the event that the patient develops these reactions, Flonoxin 400mg Tablet should be discontinued and appropriate measures instituted. Caution is recommended if Flonoxin 400mg Tablet is to be used in psychotic patients or in patients with a history of psychiatric disease

Antibiotic-associated diarrhoea including colitis
Antibiotic-associated diarrhoea (AAD) and antibiotic-associated colitis (AAC), including pseudomembranous colitis and Clostridium difficile-associated diarrhoea, has been reported in association with the use of broad spectrum antibiotics including moxifloxacin and may range in severity from mild diarrhoea to fatal colitis. Therefore, it is important to consider this diagnosis in patients who develop serious diarrhoea during or after the use of moxifloxacin. If AAD or AAC is suspected or confirmed, ongoing treatment with antibacterial agents, including moxifloxacin, should be discontinued and adequate therapeutic measures should be initiated immediately. Furthermore, appropriate infection control measures should be undertaken to reduce the risk of transmission. Drugs inhibiting peristalsis are contraindicated in patients who develop serious diarrhoea.

Patients with myasthenia gravis
Moxifloxacin should be used with caution in patients with myasthenia gravis because the symptoms can be exacerbated. Exacerbation of myasthenia gravis: Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

Tendinitis and tendon rupture
Tendinitis and tendon rupture (especially but not limited to Achilles tendon), sometimes bilateral, may occur as early as within 48 hours of starting treatment with fluoroquinolones and have been reported to occur even up to several months after discontinuation of treatment. The risk of tendinitis and tendon rupture is increased in older patients (above 60 years of age), with renal impairment, patients with solid organ transplants, and those treated concurrently with corticosteroids. Therefore, concomitant use of corticosteroids should be avoided. At the first sign of tendinitis (e.g. painful swelling, inflammation) the treatment with moxifloxacin should be discontinued and alternative treatment should be considered. The affected limb(s) should be appropriately treated (e.g. immobilization). Corticosteroids should not be used if signs of tendinopathy occur.

Patients with renal impairment
Elderly patients with renal disorders should use moxifloxacin with caution if they are unable to maintain adequate fluid intake, because dehydration may increase the risk of renal failure.

Vision disorders
If vision becomes impaired or any effects on the eyes are experienced, an eye specialist should be consulted immediately.

Dysglycemia
As with all fluoroquinolones, disturbances in blood glucose, including both hypoglycemia and hyperglycemia have been reported with moxifloxacin. In moxifloxacin-treated patients, dysglycemia occurred predominantly in elderly, diabetic patients receiving concomitant treatment with an oral hypoglycemic agent (e.g. sulfonylurea) or with insulin. In diabetic patients, careful monitoring of blood glucose is recommended.

Prevention of photosensitivity reactions
Quinolones have been shown to cause photosensitivity reactions in patients. However, studies have shown that moxifloxacin has a lower risk to induce photosensitivity. Nevertheless patients should be advised to avoid exposure to either UV irradiation or extensive and/or strong sunlight during treatment with moxifloxacin.

Patients with glucose-6-phosphate dehydrogenase deficiency
Patients with a family history of, or actual glucose-6-phosphate dehydrogenase deficiency are prone to haemolytic reactions when treated with quinolones. Therefore, moxifloxacin should be used with caution in these patients.

Patients with galactose intolerance, Lapp lactase deficiency or glucose-galactose malabsorption
Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Patients with pelvic inflammatory disease
For patients with complicated pelvic inflammatory disease (e.g. associated with a tubo-ovarian or pelvic abscess), for whom an intravenous treatment is considered necessary, treatment with moxifloxacin is not recommended. Pelvic inflammatory disease may be caused by fluoroquinolone-resistant Neisseria gonorrhoeae. Therefore in such cases empirical moxifloxacin should be co-administered with another appropriate antibiotic (e.g. a cephalosporin) unless moxifloxacin-resistant Neisseria gonorrhoeae can be excluded. If clinical improvement is not achieved after 3 days of treatment, the therapy should be reconsidered.

Interference with biological tests
Moxifloxacin therapy may interfere with the Mycobacterium spp. culture test by suppression of mycobacterial growth causing false negative results in samples taken from patients currently receiving moxifloxacin.

Patients with MRSA infections
Moxifloxacin is not recommended for the treatment of MRSA infections. In case of a suspected or confirmed infection due to MRSA, treatment with an appropriate antibacterial agent should be started.

Aortic aneurysm and dissection
Epidemiologic studies report an increased risk of aortic aneurysm and dissection after intake of fluoroquinolones, particularly in the older population. Therefore, fluoroquinolones should only be used after careful benefit-risk assessment and after consideration of other therapeutic options in patients with positive family history of aneurysm disease, or in patients diagnosed with pre-existing aortic aneurysm and/or aortic dissection, or in presence of other risk factors or conditions predisposing for aortic aneurysm and dissection (e.g. Marfan syndrome, vascular Ehlers-Danlos syndrome, Takayasu arteritis, giant cell arteritis, Behcet 's disease, hypertension, known atherosclerosis). In case of sudden abdominal, chest or back pain, patients should be advised to immediately consult a physician in an emergency department.

INTERACTIONS WITH OTHERS MEDICAMENTS:
For the following substances absence of a clinically relevant interaction with moxifloxacin was proven: atenolol, ranitidine, calcium supplements, theophylline, oral contraceptives, glibenclamide, itraconazole, digox

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Product : T FLONOXIN 400MG TAB
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Artwork Type [PLEASE TICK ✓]

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☐ Outer Box
☐ Label

☒ Leaflet
☐ Fix-A-Form
☐ Others : _____

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Sign / Stamp

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Use in Lactation: There is no data available in lactating or nursing women. Therefore, the use of moxifloxacin in nursing mothers is contraindicated.

EFFECTS ON ABILITY TO DRIVE OR USE MACHINES:
Fluoroquinolones including moxifloxacin may result in an impairment of the patient's ability to drive or operate machinery due to CNS reactions and vision disorders.

SIDE EFFECTS/ADVERSE REACTIONS:
Adverse reactions are listed below:
Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness. Frequencies are defined as:
• common
• uncommon
• rare
• very rare

System Organ Class (MedDRA)	Common	Uncommon	Rare	Very Rare
Infections and infestations	Superinfections due to resistant bacteria or fungi e.g. oral and vaginal candidiasis			
Blood and lymphatic system disorders		Anaemia Leucopenia(s) Neutropenia Thrombocytopenia Thrombocythemia Blood eosinophilia Prothrombin time prolonged /INR increased		Prothrombin level increased/INR decreased Agranulocytosis
Immune system disorders		Allergic reaction	Anaphylaxis incl. very rarely life-threatening shock Allergic oedema / angiooedema (including laryngeal oedema, potentially life-threatening)	
Metabolism and nutrition disorders		Hyperlipidemia	Hyperglycemia Hyperuricemia	Hypoglycemia
Psychiatric disorders*		Anxiety reactions Psychomotor hyperactivity/ agitation	Emotional lability Depression (in very rare cases potentially culminating in self-injurious behaviour, such as suicidal ideations/ thoughts, or suicide attempts) Hallucination	Depersonalization Psychotic reactions (potentially culminating in self-injurious behaviour, such as suicidal ideations/ thoughts, or suicide attempts)
Nervous system disorders*	Headache Dizziness	Par- and Dysaesthesia Taste disorders (incl. ageusia in very rare cases) Confusion and disorientation Sleep disorders (predominantly insomnia) Tremor Vertigo Somnolence	Hypoaesthesia Smell disorders (incl. anosmia) Abnormal dreams Disturbed coordination (incl. gait disturbances, esp. due to dizziness or vertigo) Seizures incl. grand mal convulsions Disturbed attention Speech disorders Amnesia Peripheral neuropathy and polyneuropathy	Hyperaesthesia
Eye disorders*		Visual disturbances incl. diplopia and blurred vision (especially in the course of CNS reactions)		Transient loss of vision (especially in the course of CNS reactions)
Ear and labyrinth disorders*			Tinnitus Hearing impairment incl. deafness (usually reversible)	
Cardiac disorders	QT prolongation in patients with hypokalaemia	QT prolongation Palpitations Tachycardia Atrial fibrillation Angina pectoris	Ventricular tachyarrhythmias Syncope (i.e., acute and short lasting loss of consciousness)	Unspecified arrhythmias Torsade de Pointes Cardiac arrest
Vascular disorders		Vasodilatation	Hypertension Hypotension	Vasculitis
Respiratory, thoracic and mediastinal disorders		Dyspnea (including asthmatic conditions)		
Gastrointestinal disorders	Nausea Vomiting Gastrointestinal and abdominal pains Diarrhoea	Decreased appetite and food intake Constipation Dyspepsia Flatulence Gastritis Increased amylase	Dysphagia Stomatitis Antibiotic associated colitis (incl. pseudo-membranous colitis, in very rare cases associated with life-threatening complications)	
Hepatobiliary disorders	Increase in transaminases	Hepatic impairment (including LDH increase) Increased bilirubin Increased gamma-glutamyl-transferase Increase in blood alkaline phosphatase	Jaundice Hepatitis (predominantly cholestatic)	Fulminant hepatitis potentially leading to life-threatening liver failure (including fatal cases)
Skin and subcutaneous tissue disorders		Pruritus Rash Urticaria Dry skin		Bullous skin reactions like Stevens-Johnson syndrome or toxic epidermal necrolysis (potentially life-threatening)
Musculoskeletal and connective tissue disorders*		Arthralgia Myalgia	Tendonitis Muscle cramp Muscle twitching Muscle weakness	Tendon rupture Arthritis Muscle rigidity Exacerbation of symptoms of myasthenia gravis
Renal and urinary disorders		Dehydration	Renal impairment (including increase in BUN and creatinine) Renal failure	

General disorders and administration site conditions*		Feeling unwell (predominantly asthenia or fatigue) Painful conditions (including pain in back, chest, pelvic and extremities) Sweating	Oedema	
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There have been very rare cases of the following side effects reported following treatment with other fluoroquinolones, which might possibly also occur during treatment with moxifloxacin: hypernatraemia, hypercalcaemia, haemolytic anaemia, rhabdomyolysis, photosensitivity reactions.

*Very rare cases of prolonged (up to months or years), disabling and potentially irreversible serious drug reactions affecting several, sometimes multiple, system organ classes and senses (including reactions such as tendinitis, tendon rupture, arthralgia, pain in extremities, gait disturbance, neuropathies associated with paraesthesia, depression, fatigue, memory impairment, sleep disorders, and impairment of hearing, vision, taste and smell) have been reported in association with the use of fluoroquinolones in some cases irrespective of pre-existing risk factors (see section Warnings and Precautions).

Exacerbation of Myasthenia Gravis
Post Marketing Experience.

SYMPTOMS AND TREATMENT OF OVERDOSAGE:
Only limited data on overdose are available. No specific countermeasures after accidental overdose are recommended. In the event of overdose, symptomatic treatment should be implemented. ECG monitoring should be undertaken, because of the possibility of QT interval prolongation. Concomitant administration of charcoal with a dose of 400 mg oral moxifloxacin will reduce systemic availability of the drug by more than 80%. The use of charcoal early during absorption may be useful to prevent excessive increase in the systemic exposure to moxifloxacin in cases of oral overdose.

STORAGE CONDITIONS:
Store in a dry place below 30°C.
Store in original package to protect from moisture.
Keep out of reach of children.
Jauhkan daripada kanak-kanak.

PACKING/ PACK SIZES:
Flonoxin 400mg Tablets: box of 1 x 5 tablets; Box of 2 x 5 tablets.

SHELF LIFE:
Please refer to outer package.

PRODUCT REGISTRATION HOLDER:
Duopharma Marketing Sdn. Bhd.
Lot No. 2, 4, 6, 8, & 10, Jalan P/7, Section 13, Bangi Industrial Estate,
43650 Bandar Baru Bangi, Selangor, Malaysia.

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